

Theme 1: FUNCTIONS

Note Title

2/5/2018

Unit 1.1 Numbers, Intervals and Inequalities.

- In this lecture we focus on Inequalities.
- **Read Stewart pp. A2-A6**
- How you write Mathematics is important, i.e., Notation and symbols are important.

⇒ Know how to use the imply sign

$\cap$ Intersection of 2 sets (AND)

$\cup$ Union of 2 sets (OR)

- To solve an inequality: use a graph where possible.
- To solve a quadratic inequality: ALWAYS use a graph.
- p. A4: Know the Table of Intervals and the Rules of Inequalities.

Examples:

1) Write in interval notation

a) $x > 3$ and $x < 5$

$(3, \infty) \cap (-\infty, 5)$

$x \in (-\infty, 5) \cap (3, \infty)$

$\Rightarrow x \in (3, 5)$

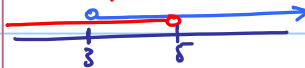


b) $x < 5$ or $x > 3$

$(-\infty, 5) \cup (3, \infty)$

$\Rightarrow x \in (-\infty, 5) \cup (3, \infty)$

$\Rightarrow \underline{x \in \mathbb{R}}$



c) $x < 3$ or $x > 5$

$(-\infty, 3) \cup (5, \infty)$

$\Rightarrow x \in (-\infty, 3) \cup (5, \infty)$



d) $x < 3$ and $x > 5$

$(-\infty, 3) \cap (5, \infty)$

$x \in (-\infty, 3) \cap (5, \infty)$

$\Rightarrow x \in \emptyset$

$\emptyset \equiv$ The empty set.



SOLVE FOR x :

2) $x(x+1) = 0$

$\Rightarrow x = 0$ or $x+1 = 0$

$\Rightarrow x = 0$ or $x = -1$

$\Rightarrow x \in \{-1, 0\}$

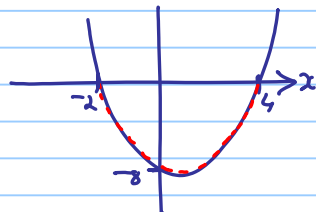
3) $x^2 < 2x+8$

$x^2 - 2x - 8 < 0$ (Always use a graph!)

$\Rightarrow (x-4)(x+2) < 0$

The parabola is negative between -2 and 4,

$\therefore x \in (-2, 4)$



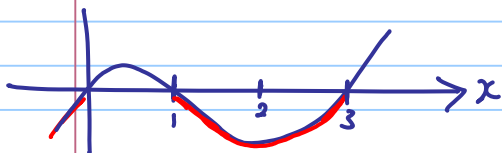
$$(3) x^3 + 3x < 4x^2$$

$$x^3 - 4x^2 + 3x < 0$$

$$\Rightarrow x(x^2 - 4x + 3) < 0$$

$$\Rightarrow x(x-3)(x-1) < 0$$

From this information we learn that we have a third degree polynomial with zeros at $x=0, 1$ and 3 .



When $x=2$, we have

$$2(2-3)(2-1)$$

$$= 2(-1)(1)$$

$$= -2.$$

This gives us the form of the polynomial.

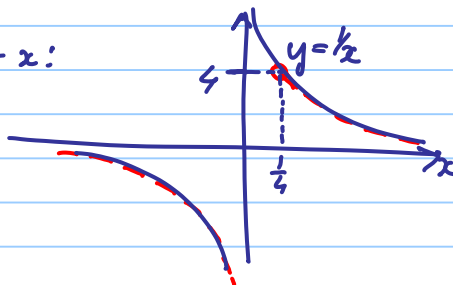
$$\Rightarrow x \in (-\infty, 0) \cup (1, 3),$$

since on these intervals, the polynomial is negative.

$$(4) \frac{1}{x} < 4.$$

We use a graph to solve for x :

$\Rightarrow$ According to the graph, $\frac{1}{x} < 4$ if $x \in (-\infty, 0) \cup (\frac{1}{4}, \infty)$.



$$(5) \frac{x+1}{2-x} \geq 0$$

$$\frac{x+1}{2-x} (2-x)^2 \geq 0 \times (2-x)^2$$

$$, \text{ NB } x-2 \neq 0.$$

$$\Rightarrow (x+1)(2-x) \geq 0$$

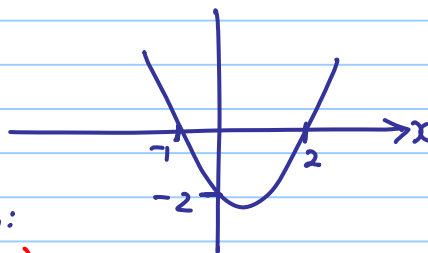
$$\Rightarrow -(x+1)(x-2) \geq 0$$

$$\Rightarrow (x+1)(x-2) \leq 0$$

$$\Rightarrow x^2 - x - 2 \leq 0$$

Now we use the graph to solve the inequality:

$$\Rightarrow x \in [-1, 2) \quad (x \neq 2)$$



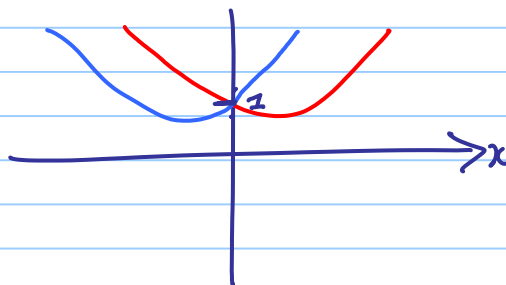
$$(6) -x^2 - x - 1 > 0$$

$$\Rightarrow x^2 + x + 1 < 0$$

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= \frac{-1 \pm \sqrt{-3}}{2}$$



$\Rightarrow$ The equation does not have any real solutions.

$$\Rightarrow x \in \emptyset.$$

